

(4) C++ Abstractions

Nico Ludwig (@ersatzteilchen)

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- Sources:
 - Bjarne Stroustrup, The C++ Programming Language
 - John Lakos, Large-Scale C++ Software Design

Initial Words

Yes, my slides are heavy.

I do so, because I want people to go through the slides at their own pace w/o having to watch an accompanying video.

On each slide you'll find the crucial information. In the notes to each slide you'll find more details and related information, which would be part of the talk I gave.

Have fun!

Cstrings revisited

- In the last lectures we talked about the encapsulation of concepts and RAII.
 - Now it's time to make use of RAII to make our programming tasks easier.
- E.g. we should apply RAII to encapsulate cstrings, thus enhancing:
 - creation,
 - assignment,
 - copying and
 - operations on cstrings
 - by encapsulating the tedious memory management around cstrings.
- The good news: we don't have to create a new UDT, we can use STL-strings!
 - STL-strings are represented by the C++ standard type `std::string`.
 - The type `std::string` is defined in `<string>`.
 - `std::string` provides important RAII features that make using `std::string` intuitive.

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- Also the ugly 0-termination is to be encapsulated.
- The type `std::string` is a very good example of encapsulation. The design of `std::string` makes working with strings very comfortable, safe and productive, esp. because RAII has been implemented in `std::string`.
- Many different libraries implemented "their" string types different from `std::string` (mutable string types, immutable string types, reference counted string types etc.), but we'll only use `const char*` or `std::string` in this course.

The UDT `std::string`

- To use STL-strings we have to `#include <string>` and keep on going with `std::string`:

```
std::string name = "Lola"; // Creation
std::string otherName = name; // Assignment/Copying
std::string subString = name.substr(2, 2); // Substring: a member function of std::string (result: "la")
// The memory of the used strings will be managed (e.g. also freed) by RAII.
```

- The type `std::string` can be used like a fundamental type, e.g. as `return`/parameter type:

```
// Using std::string as return and parameter type:
std::string AcceptsAndReturnsSTLString(std::string name) {
    std::cout<<"The passed name: "<<name<<std::endl;
    // Create and return an STL-string from cstring literal:
    return "Pamela";
}
```

```
// Create and accept STL-string from a cstring literal:
std::string result = AcceptsAndReturnsSTLString("Sandra");
// > The passed name: Sandra
// result = "Pamela"
```



- Hooray! – A simple-to-use string type in C++! But there is a performance problem:
 - Due to RAII, `std::strings` are copied when passed to and returned from functions.
 - It means that the encapsulated cstrings (`char` arrays) are allocated and freed multiply.
 - This is, e.g., done by ctors and dtors which manage RAII.
 - Can we improve this situation?

Avoiding Object Copies – Example: std::string

- Means to avoid multiple copying of *std::strings*:
 - Pass pointers to *std::strings* to functions (i.e. back to the idea "pass by reference").
 - Return pointers to *std::strings* created in the heap or freestore.
 - ... let's rewrite *AcceptsAndReturnsSTLString()* accordingly:

```
// Using std::string* as return and parameter type:
std::string* AcceptsAndReturnsSTLString(std::string* name) {
    std::cout<<"The passed name: "<<*name<<std::endl;
    // Create std::string on freestore and return it.
    return new std::string("Pamela");
}
```

```
// Create std::string on the stack and pass a pointer to that
// string to the function:
std::string sandra = "Sandra";
std::string* result = AcceptsAndReturnsSTLString(&sandra);
// > The passed name: Sandra
// result = pointer to "Pamela"
delete result;
```



- The means work: neither *sandra* nor *"Pamela"* is copied, but created only once.
- But dealing with freestore and pointers to avoid copies is risky and cumbersome.
 - Therefore C++ introduced another means to avoid copies: references.

C++ References

- We can, e.g., use std::string-references as parameter types.
 - (For the time being, we'll only discuss parameters of reference type.)

```
// Using an std::string-reference as parameter type:  
void AcceptsSTLString(std::string& name) {  
    std::cout<<"The passed name:"<<name<<std::endl;  
}
```

```
// Create std::string on stack and pass it to the function:  
std::string pamela = "Pamela";  
AcceptsSTLString(pamela); // Here: No extra syntax!
```

```
// Questionable! Field as C++ reference.  
class Foo {  
    int& refToInt;  
public  
    // The field refToInt needs to be initialized:  
    Foo(int i) : refToInt(i){} // Initializer list needed!  
};
```

```
// Questionable! Local variable as C++ reference.  
int i = 23;  
// The reference refToInt needs to be initialized:  
int& refToInt = i;
```

- Syntactic peculiarities of C++ references:
 - A C++ reference has a syntax decoration of the referenced type with the &-symbol.
 - (+) References cannot be uninitialized and cannot be 0.
 - (+) When an argument is passed to a reference parameter, no extra syntax is involved.
 - Using reference parameters leads to more unobtrusive code than with pointer parameters.
 - (+) References can also be used as local variables and fields, but it leads to questionable code.
 - (-) References need to be initialized in opposite to pointers!
 - (-) References do have no notion of "nullity" like pointers (pointers can be 0).
 - (-) Functions differing only in the "reference-ness" of its parameters do not overload.

Superficial Definition

A C++ reference is a kind of pointer, that cannot be 0 or invalid/uninitialized.

Tracing Object Lifetime

```
class PersonLitmus { // Shows, when an instance is
public:               // created and destroyed.
    PersonLitmus() {
        std::cout<<"Person created"<<std::endl;
    }
    ~PersonLitmus() {
        std::cout<<"Person destroyed"<<std::endl;
    }
};
```

- Let's reuse the type *PersonLitmus* with tracing messages.

- Calling the function accepting *PersonLitmus* leads to anonymous copies:

```
void AcceptsPerson(PersonLitmus person) {
    // pass
}
```

```
PersonLitmus person;
// >Person created
AcceptsPerson(person);
// >Person destroyed (destroys anonymous copy)
// >Person destroyed (destroys person)
```

- Calling the function accepting *PersonLitmus&* avoids anonymous copies:

```
void AcceptsPersonByRef(PersonLitmus& person) {
    // pass
}
```

```
PersonLitmus person;
// >Person created
AcceptsPersonByRef(person);
// >Person destroyed (destroys person)
```

Anonymous Object Copies

- Let's review the example without reference parameters:

```
void AcceptsPerson(PersonLitmus person) {  
    // pass  
}
```

```
PersonLitmus person;  
// >Person created  
AcceptsPerson(person);  
// >Person destroyed (destroys anonymous copy)  
// >Person destroyed (destroys person)
```



- This example shows that more objects seem to be destroyed than created!
 - What the heck is going on here? The ctor was only called once, but the dtor twice!
- By default, C++ passes/returns objects to/from functions by value, creating copies.
- The answer for these unbalanced dtors: temporary copies of those objects are created.
- It's time to clarify the source of the anonymous copies: copy constructors (cctors).
 - Cctor are called when objects are getting copied!

Tracing Object Copies

```
class PersonLitmus {           // Shows when an instance is
public:                         // created, copied and destroyed.
    PersonLitmus() {
        std::cout<<"Person created ("<<this<<")"<<std::endl;
    }
    PersonLitmus(PersonLitmus& original) {
        std::cout<<"Person copied ("<<this<<")"<<std::endl;
    }
    ~PersonLitmus() {
        std::cout<<"Person destroyed ("<<this<<")"<<std::endl;
    }
};
```

```
void AcceptsPerson(PersonLitmus person) {
    // pass
}
```

```
PersonLitmus person;
// >Person created (0x7ffefbfff2b8)
AcceptsPerson(person);
// >Person copied (0x7ffefbfff2b0) (anon. copy created)
// >Person destroyed (0x7ffefbfff2b0) (destroys anon. copy)
// >Person destroyed (0x7ffefbfff2b8) (destroys person)
```

- Implementing the copy constructor with tracing makes the creation of copies visible.
 - If not explicitly programmed, a `public` ctor will be generated automatically (like dtors/dtors are).
- Syntactic peculiarities of ctors:
 - It has exactly one (non-defaulted) parameter: a reference of the surrounding type.
 - The ctor obeys to the same syntactic rules like other ctors do.
 - A type's ctor will be automatically called, when the type is passed/returned by value.

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- Why needs a ctor to accept a reference of the surrounding type?
 - If the ctor accepted a non-reference parameter of the surrounding type, the ctor would call itself recursively up to infinity!
- Cctors' parameters are allowed to be annotated with all kinds of cv-qualified references, they also overload (That should not be done on purpose!). – The rules for matching overloads are complex and will not be discussed here.
- Cctors can have more than the obligatory parameter, but the remaining parameters need to be notated with default arguments.

The Problem with shared Data – Copying

```
class Person { // (members hidden)
    char* name;
public:
    Person(const char* name) {
        if (name) {
            this->name = new char[std::strlen(name) + 1];
            std::strcpy(this->name, name);
        }
    }
    ~Person() {
        delete[] this->name;
    }
};
```

```
void AcceptsPerson(Person person) {
    // pass
}
```

```
Person nico("nico");
AcceptsPerson(nico);
// Undefined behavior: crash!
```

// Virtually something like this is executed:

```
Person nico("nico");
Person tmp;
tmp.name(nico.name); // Copies the pointer.
AcceptsPerson(tmp);
```

- Let's revisit the type *Person* that handles dynamic data.
 - Passing an object of type *Person* by value leads to undefined behavior in the dtor!
- The automatically created ctor just copies all fields, not the referenced memory.
 - It results in *Person*-copies having copied pointers to the same location in memory (*name*).
 - Every *Person*-copy will try to delete (dtor) the same location in memory via its pointer-copy.

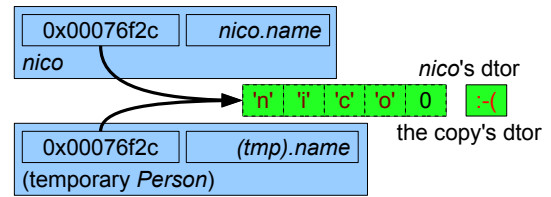
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- The syntax *tmp.name(nico.name)* is not legal, a ctor can't be called like so. – But this is what's happening underneath basically.

The automatically created Copy Constructor

```
class Person { // (members hidden)
    char* name;
};

{
    Person nico("nico");
    AcceptsPerson(nico);
} // Undefined behavior when nico's scope ends!
```



- When `nico` is passed to `AcceptsPerson()` a copy of `nico` is created.
 - The automatically generated ctor does only copy `nico`'s fields (i.e. `name`).
 - The ctor doesn't copy the occupied memory in depth, we call this a shallow copy.
 - The automatically generated ctor is **public** also for **classes**!
- We've two `Persons`' `name`-fields both pointing to the same location in the freestore.
 - This leads to dtors of two `Persons` feeling responsible for `name`'s memory.
 - => In effect the same memory will be freed twice!
 - We hurt a fundamental rule when dealing w/ dynamic memory: We should not free dynamically created content more than once!

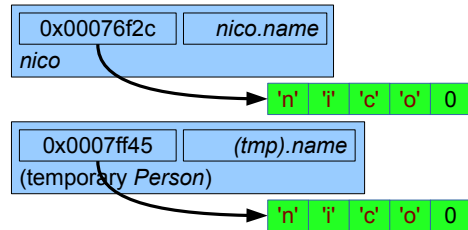
The Copy Constructor

```
class Person { // (members hidden)
    char* name;
public:
    Person(const char* name) {
        if (name) {
            this->name = new char[std::strlen(name) + 1];
            std::strcpy(this->name, name);
        }
    }
    Person(Person& original) { // The cctor.
        name = new char[std::strlen(original.name) + 1];
        std::strcpy(name, original.name);
    }
    ~Person() {
        delete[] this->name;
    }
};
```

- The solution: we've to implement the cctor explicitly!
 - We have to implement the cctor to make a deep copy of the original.
 - The cctor just accepts the original object and makes a deep copy.
 - std::string provides a cctor.

```
void AcceptsPerson(Person person) {
    // pass
}
```

```
{
    Person nico("nico");
    AcceptsPerson(nico);
} // Fine!
```



Implicit and Explicit Conversion Constructors

- In fact we have already overloaded an operator in the UDT *Person*!
 - Every single-parameter ctor is also an implicit conversion ctor.
 - E.g. a `const char*` passed to *AcceptsPerson()* will be implicitly converted to a *Person*.
 - Also `std::string` has such an implicit conversion ctor: `string::string(const char*)`.

```
class Person { // (members hidden)
public:
    Person(const char* name) {
        // pass
    }
};
```

```
void AcceptsPerson(Person person) {
    // pass
}
```

```
AcceptsPerson("nico"); // Ok!
```

- Sometimes implicit conversion is not desired for single-parameter ctors.
 - (E.g. to avoid "surprises" with overloaded functions.)
 - Therefore C++ allows conversion ctors to be marked as explicit conversion ctors.

```
class Person { // (members hidden)
public:
    explicit Person(const char* name) {
        // pass
    }
};
```

```
void AcceptsPerson(Person person) {
    // pass
}
```

```
AcceptsPerson("nico"); // Invalid! No implicit conversion!
```

```
AcceptsPerson(Person("nico")); // Ok! Explicit conversion.
```

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- HandlePerson(Person)* and *HandlePerson(const Person&)* would not be called, if *HandlePerson(const char*)* existed and *HandlePerson("nico")* be called. The best match is taken, implicit conversion and temporary copies are avoided at overload resolution.

Problems with Reference Parameters

- After the discussion concerning costly copying, let's use references for all UDTs to avoid (deep) copying in future! – But there are problems:

- 1. With functions accepting references we could modify the original argument!

- Very often this happens accidentally, here it is shown in a more radical example:

```
void PrintToConsole(std::string& text) {  
    std::cout<<text<<std::endl;  
    // Oops! Modified the parameter and the argument!  
    text = "Angela";  
}
```

```
std::string pamela = "Pamela";  
PrintToConsole(pamela);  
// >Pamela  
std::cout<<pamela<<std::endl;  
// >Angela (Oops! The content of pamela has been modified!)
```

- 2. Functions accepting references can't cope with objects created by implicit conversions.

- E.g. a `const char*` can't be implicitly converted into an `std::string` and passed to an `std::string&`:

```
void PrintToConsole(std::string& text) {  
    std::cout<<text<<std::endl;  
}
```

```
// Invalid! Literal const char* can't be passed!  
PrintToConsole("Pamela");
```

- This yields a compiler message like "Non l-value can't be bound to non-`const` reference."

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- An object created by implicit conversion is a temporary object, and as such it can't be an lvalue, therefor this message happens to appear.

Const References

- The discussed problems can be solved with [const](#) references ([const&](#)).

```
void PrintToConsole(const std::string& text) {  
    std::cout<<text<<std::endl;  
    // Invalid! Const parameter.  
    text = "Angela";  
}
```

```
std::string pamela = "Pamela";  
PrintToConsole(pamela);  
// >Pamela
```

- [const&](#) parameters cannot be modified (accidentally), e.g. assigned to.
 - [const std::string&](#) are esp. important as [std::string](#) is a mutable string type.
 - The same is valid for [const](#) pointers (e.g. also as parameters).
- [const&](#) parameters can accept temporary anonymous conversion copies.
 - A [const char*](#) can be implicitly converted into an [std::string](#) and passed to a [const std::string&](#):

```
void PrintToConsole(const std::string& text) {  
    std::cout<<text<<std::endl;  
}
```

```
// Fine! Literal const char* can be passed now!  
PrintToConsole("Pamela");
```

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- This is a very important fact: [const char*](#) is non-modifiable, but the STL type [std::string](#) is mutable! With [const std::strings](#), (and [const&](#) thereof) we can get [const char*](#)'s safety aspect back.
- Strings in Java and .NET are not mutable. On that platforms string operations create new strings instead of modifying the string on which they have been called.

The need for const Member Functions

- OK! Then let's use `const&` for all UDTs! – But there are still problems:
 - We can't call all the member functions via a `const&`.

```
class Date { // (members hidden)
    int month;
public:
    int GetMonth() {
        return month;
    }
};
```

```
void PrintMonth(const Date& date) {
    // Invalid! Can't call non-const member function on const&.
    std::cout<<date.GetMonth()<<std::endl;
}
```

```
void PrintMonth(const Date* date) {
    // Invalid! Can't call non-const member function on const pointer.
    std::cout<<date->GetMonth()<<std::endl;
}
```

- C++ assumes that all member functions potentially modify the target object.
 - Calling member functions on `const&` and `const` pointers is generally not allowed.
 - Compilers can't predict, whether called member functions modify the object.
- To solve this problem C++ introduces so called `const` member functions.
 - Only `const` member functions can be called on `const` objects.

Implementing const Member Functions

- To make a member function **const**: just add the **const** suffix to declaration and definition.

```
class Date { // (members hidden)
public:
    int month;
    int GetMonth() const {
        return month;
    }
};
```

```
void PrintMonth(const Date& date) {
    // Fine! Date::GetMonth() is a const member function.
    std::cout<<date.GetMonth()<<std::endl;
}
```

```
void PrintMonth(const Date* date) {
    // Fine! Date::GetMonth() is a const member function.
    std::cout<<date->GetMonth()<<std::endl;
}
```

- In the implementation of a **const** member function
 - fields of the enclosing UDT/instance can only be read.
 - only **static** member functions and other **const** member functions of the enclosing UDT/instance can be called.
 - The **const**-ness of **const** member functions is kind of "closed".
 - `std::string` provides a set of **const** member functions.
- Member functions and **const/volatile** qualifiers (cv-qualifiers):
 - We can have a **const** and a non-**const** overload of a member function in C++.
 - C++ also allows the definition of **volatile** and **const volatile** member functions.
 - Non-**const**, **const**, **volatile** and **const volatile** do overload!

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- What is "volatile"?
 - If an object is declared **volatile**, the C/C++ compiler assumes that its value could change anytime by any thread of execution. – Accessing a **volatile** variable is interpreted as direct memory read/write without caching.
 - The cv-qualification needs to be repeated on a non-inline definition, because cv-qualifiers overload.
 - There can be no **const static** member functions.

C++ References summarized: Our Rules for References

- C++ references have been introduced to overload operators, because:
 - they provide an unobtrusive syntax to make using operators intuitively,
 - they prevent call by value and
 - const& allow passing temporary objects, so its also good for non-operator functions.
- Important features of C++ references:
 - They're similar to pointers as call by reference, aliasing and data sharing is concerned.
 - Derived constraints because references are not having a memory or "location" manifestation:
 - We cannot have pointers to references, arrays of references and references to references.
 - References can't be uninitialized and they have no notion of "nullity".
- Some widely used industry standards that we're going to adopt as rules:
 - (+) Primary use const& for UDT parameters, and avoid passing UDTs by value!
 - UDTs should be passed by const&, if they are not already being passed as const pointer.
 - (+) Pass fundamental types only by value or pointer, the compiler cares for reasonable optimization.
 - (+) If we need to modify passed objects we should use non-const pointers.
 - (-) Don't use references to replace pointers only for syntax purposes in order to modify the passed objects!

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- If we see the discussed rules hurt, then this is what we call a "code smell". If the code is under our control, we should refactor it to obey these rules and prove the functionality with unit tests.

Circumventing const: mutable Fields

- We can't modify the target object (i.e. `this`) in `const` member functions.

- We can not write a `const Date::SetMonth()` member function like so:

```
class Date { // (members hidden)
    int month;
public:
    void SetMonth(int month) const {
        this->month = month; // Invalid! this->month is readonly
                             // in a const member function.
    }
};
```

- Often real life objects need to differentiate logical from physical `const`-ness.
 - I.e. an object presents `const` member functions that need to write some fields.
 - C++ provides a backdoor: `mutable` fields can be written in `const` member functions.

```
class Date { // (members hidden)
    int month;
    mutable int monthAccessed;
public:
    int GetMonth() const {
        ++monthAccessed; // Fine! Mutable fields can be written
        return month;    // in const member functions.
    }
};
```

Using mutable Fields to cope with lags in const-ness

- Assume an earlier version of *Date*, where *GetMonth()* was a non-const member function:

```
class Date { // (members hidden)
    int month;
public:
    int GetMonth() { // non-const!
        return month;
    }
};
```

- But in this state we cannot call non-const *Date::GetMonth()* in a const context like the const member function *Person::GetMonthOfBirth()*.

```
class Person { // (members hidden)
    Date date;
public:
    int GetMonthOfBirth() const {
        return date.GetMonth();
    }
};
```

- For the moment let's further assume we do not own *Date*'s code, i.e. we can't change *Date* and make *Date::GetMonth()* const.

- To solve the issue we can make *Person::date* a mutable field.
- We can use mutable to deal with fields, whose type does not (yet) implement const-ness "consequently".

```
class Person { // (members hidden)
    mutable Date date;
public:
    int GetMonthOfBirth() const {
        return date.GetMonth();
    }
};
```

Cheating const: Casting const-ness away

- After discussing how to avoid copies, implicit conversion and `std::string`, we'd better use `const std::string&` as function parameter type always.

- But this also means that passed `std::strings` can't be modified!

```
void Clear(const std::string& name) {  
    name.erase(); // Invalid! Can't call non-const member function on name.  
} // That makes sense! std::string::erase() can't be a const member function!
```

- There is a way to call non-const member functions on const objects!

- `const`-ness can be casted away in C++ with the `const_cast` operator:

```
void ClearConst(const std::string& name) {  
    std::string& nonConstName = const_cast<std::string&>(name);  
    nonConstName.erase(); // Ok! nonConstName is not const!  
}
```

- Casting `const`-ness away is dangerous! – It's undefined with temporary objects.

```
std::string nico("nico")  
ClearConst(nico); // Ok!  
std::cout<<nico<<std::endl;  
// > // (empty)
```

```
ClearConst("nico"); // Undefined behavior! The literal "nico"  
// is really const and can't be erased!  
std::cout<<nico<<std::endl;  
// > // (empty)
```

- Let's never use `const_cast`, unless we've a very good reason to use it!

"Const-incorrectness"

- All fields of a `const` object are also `const`, we can only call `const` member functions on it.

- Assume the UDT *Person* and the `const` member function *GetName()* in (1):

```
class Person { // (1) (members hidden)
public:
    char* name; // Only for demo!
    char* GetName() const {
        return this->name;
    }
};
```

```
class Person { // (2) (members hidden)
public:
    char* const name;
    char* GetName() const {
        return this->name;
    }
};
```

- When we define a `const` instance of *Person*, the UDT virtually changes to (2).

```
// We already know this fact:
const Person nico("nico");
// This is not allowed:
nico.name = 0; // Invalid!
// The field nico.name is a const pointer!
```

```
const Person nico("nico"); // But here the surprises:
// These operations are allowed(!):
std::strcpy(nico.name, "joe");
// The memory to which nico.name points to is not const!
std::strcpy(nico.GetName(), "jim");
// The memory to which nico.name points to is not const!
// - GetName() just returns a pointer.
```

- Was all our work for `const`-ness for good? – Well, our UDT isn't yet `const`-correct!

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- Another way to understand `const` correctness: `const` member functions need to be balanced to other `const` member functions and their return types. Finally, `const` correct types allow the creation of immutable types in C++. (The support for immutable types is not so good in Java and C# (2018).)

Const-correctness

- The just encountered problem is that const-ness is not deep enough!
 - The const-ness of an object only affects the fields, not the memory the fields refer to.
 - This problem is relevant for pointer-fields as well as for reference-fields.
 - const member functions don't shield us from modifiable referred memory!
- To make *Person/GetName()* const-correct, we'll make the field *name* private and fix *GetName()* to return a `const char*` instead of a `char*`.
 - 1. The field *name* can't be accessed from "outside".
 - 2. *GetName()* returns a `const char*`, it doesn't allow writing memory referenced by *name*.

```
class Person { // (members hidden)
    char* name;
public:
    const char* GetName() const {
        return this->name;
    }
};
```

```
const Person nico("nico"); // Now Person is const correct.
```

```
std::strcpy(nico.name, "joe"); // Invalid! name is private!
```

```
std::strcpy(nico.GetName(), "jim"); // Invalid! GetName()
// returns a const char* that doesn't allow to write the
// referred memory.
```

- const correct: Generally return const pointers/references from const member function.

Thank you!